

Principles of Economics

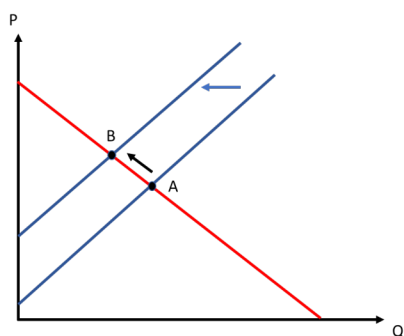
Problem Set 6

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. We have used chocolate bars as a hypothetical example of an approximately competitive market. But in recent years, producers of best-selling chocolate bars worldwide have been accused of colluding with each other to keep prices high. In what ways does the market for chocolate bars fail to satisfy the conditions for perfect competition?
2. Each brand of chocolate bar faces competition from many other similar brands. Why, despite this, do some producers have considerable market power?
3. Use the supply-and-demand model to represent the following events. In each case, indicate which curve(s) shift and explain the result.
 - a) The increase in the price of US raw cotton (show the market for US raw cotton, a market with many producers and buyers).
 - b) The increase in the price of Indian cotton (show the market for Indian raw cotton, a market with many producers and buyers).
 - c) The reduction in textile output in an English textile mill (show a single firm in a competitive product market).

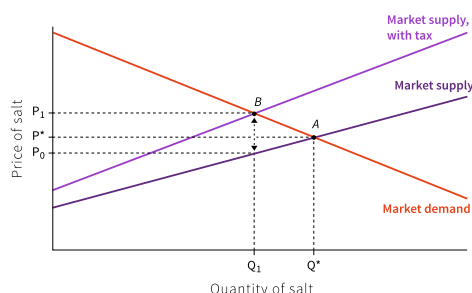


4. Consider the market for quinoa and analyze shifts in demand and supply.
 - a) In the early 2000s, there was an unexpected increase in demand for quinoa (a shift in the demand curve). What would you expect to happen to the price and quantity initially?
 - b) When demand continued to rise over the next few years, how do you think farmers responded?

- c) Why do you think the price stayed constant until 2007?
- d) Suggest an explanation for the rapid price rise in 2008 and 2009.
- e) Explain why you would or would not expect the price to fall eventually to its original level.

2. MULTIPLE CHOICE QUESTIONS

5. The following diagram depicts the demand and supply curves in the salt market. It also depicts the shift in the supply curve due to a 30% tax on the price of salt. Now suppose that the market demand curve for salt is less elastic than in the diagram. The equilibrium before tax is still at (Q^*, P^*) . Let the post-tax equilibrium be denoted by (Q_2, P_2) (not shown on the diagram). Based on this information, which of the following statements is correct?
- a. For the prices, $P^* < P_2 < P_1$.
 - b. For the outputs, $Q_1 < Q_2 < Q^*$.
 - c. The tax revenue raised would be lower if the demand curve were less elastic.
 - d. Proportionally, more of the burden of the tax would be on the producer if the demand curve were less elastic.



6. The market demand curve of a particular good is downward-sloping. Based on this information, which of the following statements is correct regarding a price-taking firm producing that good?
- a. The demand curve faced by the firm is downward-sloping.
 - b. The firm chooses the price that equals its marginal cost.
 - c. The firm chooses its output such that the marginal cost equals the price.
 - d. A price-taking firm cannot be profit-maximizing.

3. NUMERICAL PROBLEMS

7. In a perfectly competitive market, the demand is $Q = 27 - 2P$ and the supply is $Q = P - 3$. Please answer this question graphically and also analytically whenever possible.
- a. What is the equilibrium price, quantity, producer and consumer surplus in equilibrium?

- b. Suppose the government announces a tax of 3 per unit of the good on the producers. What is the new equilibrium price, quantity, producer and consumer surplus, government revenue and deadweight loss?
- c. Suppose the income of the consumers in this economy increases, and that the good is a normal good. What will happen with the equilibrium price, quantity and producer and consumer surplus? Who will benefit or lose more from the changes, and what does this depend on?

8. Answer the following questions.

- a. Sketch a diagram to illustrate the competitive market for bread, showing the equilibrium where 5,000 loaves are sold at a price of €2.00.
- b. Suppose that the bakeries get together to form a cartel. They agree to raise the price to €2.70, and jointly cut production to supply the number of loaves that consumers demand at that price. Shade the areas on your diagram to show the consumer surplus, producer surplus, and deadweight loss caused by the cartel.
- c. For what kinds of goods would you expect the supply curve to be highly elastic?
- d. Draw diagrams to illustrate how the share of the gains from trade obtained by producers depends on the elasticity of the supply curve.

4. CONCEPTUAL QUESTIONS

- 9. Consider three possible methods to sell a car that you own: a) Advertise it in the local newspaper. b) Take it to a car auction. c) Offer it to a second-hand car dealer. Would your reservation price be the same in each case? Why?
- 10. Which of the methods described in the previous question do you think would result in the highest sale price? If you used the first method, would you advertise it at your reservation price?